

CHE 311 HOMEWORK SET #1

P. 814 Ex. 1

$$a) \sum_{k=1}^3 k^2 = 1^2 + 2^2 + 3^2 = 14$$

$$b) \sum_{k=1}^3 a_k^2 = a_1^2 + a_2^2 + a_3^2$$

$$c) \sum_{j=1}^3 \sum_{k=1}^3 a_{jk} b_{jk} = (a_{1,1} \cdot b_{1,1}) + (a_{1,2} \cdot b_{1,2}) + (a_{1,3} \cdot b_{1,3}) + (a_{2,1} \cdot b_{2,1}) + (a_{2,2} \cdot b_{2,2}) + (a_{2,3} \cdot b_{2,3}) + (a_{3,1} \cdot b_{3,1}) + (a_{3,2} \cdot b_{3,2}) + (a_{3,3} \cdot b_{3,3})$$

$$d) \left(\sum_{j=1}^3 a_j \right)^2 = \sum_{j=1}^3 \sum_{k=1}^3 a_j a_k = (a_1 \cdot a_1) + (a_1 \cdot a_2) + (a_1 \cdot a_3) + (a_2 \cdot a_1) + (a_2 \cdot a_2) + (a_2 \cdot a_3) + (a_3 \cdot a_1) + (a_3 \cdot a_2) + (a_3 \cdot a_3)$$

$$= a_1^2 + 2(a_1 \cdot a_2) + 2(a_1 \cdot a_3) + a_2^2 + 2(a_2 \cdot a_3) + a_3^2$$

P. 814 Ex. 2

$$a) [v_x, v_y, v_z] \begin{bmatrix} w_x \\ w_y \\ w_z \end{bmatrix} = v_x w_x + v_y w_y + v_z w_z = 3 + -2 + -5 = \boxed{-4}$$

$$b) \begin{bmatrix} 1 & 2 & -5 \\ 3 & -1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & -5 \\ 3 & -1 & 1 \end{bmatrix} = \boxed{(-4, -16, 5)}$$

$$c) \sqrt{1^2 + 2^2 + (-5)^2} = \sqrt{1 + 4 + 25} = \boxed{\sqrt{30}}$$

$$d) (s, v) = [1, 0, 0] \begin{bmatrix} 1 \\ 2 \\ -5 \end{bmatrix} = 1 + 0 + 0 = \boxed{1}$$

$$e) \begin{bmatrix} 5 & -1 & 1 \\ 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 5 & -1 & 1 \\ 1 & 0 & 0 \end{bmatrix} = (0, 1, 1)$$

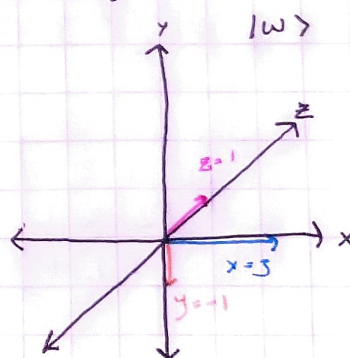
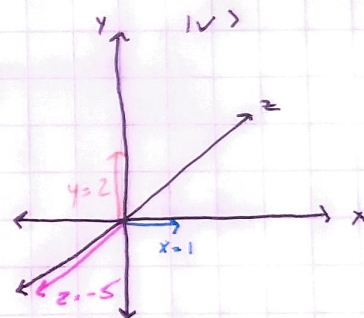
$$f) \phi_{vw} = \frac{1}{\cos \theta} = \cos^{-1} \left(\frac{v \cdot w}{\|v\| \|w\|} \right) = \cos^{-1} \left(\frac{-4}{\sqrt{30} \sqrt{1+1+1}} \right)$$

$$= \cos^{-1} \frac{-4}{\sqrt{30} \sqrt{3}}$$

$$= \cos^{-1} (-4/18.17)$$

$$= \boxed{102.72^\circ}$$

$$g) \begin{bmatrix} 1 & 2 & -5 \\ r_x & r_y & r_z \end{bmatrix} \begin{bmatrix} 1 & 2 & -5 \\ r_x & r_y & r_z \end{bmatrix} = \boxed{(2r_z - 5r_y, -5r_x - r_z, r_y - 2r_x)}$$



p. 819 Ex 1

a.)

$$\begin{bmatrix} 3 & 2 & -1 \\ 2 & 2 & 1 \\ -1 & 1 & 4 \end{bmatrix} \begin{bmatrix} 5 & 3 & -2 \end{bmatrix} = \begin{bmatrix} 15 + 6 + 2 \\ 10 + 6 + -2 \\ -5 + 3 + -8 \end{bmatrix} = \begin{bmatrix} 23 \\ 14 \\ -10 \end{bmatrix} \begin{matrix} x \\ y \\ z \end{matrix}$$

$$b.) \begin{bmatrix} 5 & 3 & -2 \end{bmatrix} \begin{bmatrix} 3 & 2 & -1 \\ 2 & 2 & 1 \\ -1 & 1 & 4 \end{bmatrix} = \begin{bmatrix} \end{bmatrix}$$

$$\begin{bmatrix} 3 & 2 & -1 \\ 2 & 2 & 1 \\ -1 & 1 & 4 \end{bmatrix} \begin{bmatrix} 5 & 3 & -2 \end{bmatrix} = \begin{bmatrix} 15 + 6 + 2 \\ 10 + 6 + -2 \\ -5 + 3 + -8 \end{bmatrix} = \begin{bmatrix} 23 \\ 14 \\ -10 \end{bmatrix} \begin{matrix} x \\ y \\ z \end{matrix}$$

$$c.) \begin{bmatrix} 3 & 2 & -1 \\ 2 & 2 & 1 \\ -1 & 1 & 4 \end{bmatrix} \begin{bmatrix} 3 & 2 & -1 \\ 2 & 2 & 1 \\ -1 & 1 & 4 \end{bmatrix} = 9 + 4 + 1 + 4 + 4 + 1 + 1 + 1 + 16 = 41$$

d.) use a answer for v.t

$$\begin{bmatrix} 5 & 3 & -2 \end{bmatrix} \begin{bmatrix} 23 \\ 14 \\ -10 \end{bmatrix} = 115 + 42 + 20 = 177$$

$$e.) \begin{bmatrix} 5 & 3 & -2 \end{bmatrix} \begin{bmatrix} 5 \\ 3 \\ -2 \end{bmatrix} = 25 + 9 + 4 = 38$$

$$f.) \begin{bmatrix} 3 & 2 & -1 \\ 2 & 2 & 1 \\ -1 & 1 & 4 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 3 & 0 & 0 \\ 2 & 0 & 0 \\ -1 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 3, 2, -1 \end{bmatrix}$$

P. 819

a) $[\delta_1 \delta_2 \cdot \delta_3] \times \delta_1 = \delta_1 \delta_2 \cos \theta \times \delta_1$

b) $\delta \cdot (\delta_1 \times \delta_2)$

c) δ^2

d) δ^2

P. 823 #5

	$\nabla \cdot \mathbf{v}$	$\nabla \mathbf{v}$	$\nabla \cdot \mathbf{vv}$
a)	$\frac{\partial b_y}{\partial x} \hat{i}$		
b)	$\frac{\partial b_x}{\partial x} \hat{i}$		
c)	$\frac{\partial b_y}{\partial x} \hat{i} + \frac{\partial b_x}{\partial y} \hat{j}$		
d)	$\frac{\partial b_y}{\partial x} \hat{i} + \frac{\partial b_x}{\partial y} \hat{j}$		

P. 823 #3

	$\nabla \cdot \mathbf{v}$	$\nabla \mathbf{v}$	$\nabla \cdot \mathbf{vv}$
a)	$\left[\frac{\partial}{\partial x} \hat{i}, \frac{\partial}{\partial y} \hat{j}, \frac{\partial}{\partial z} \hat{k} \right] \begin{bmatrix} b_y \\ 0 \\ 0 \end{bmatrix} = \frac{\partial b_y}{\partial x} = 0$		
b)	$\left[\frac{\partial}{\partial x} \hat{i}, \frac{\partial}{\partial y} \hat{j}, \frac{\partial}{\partial z} \hat{k} \right] \begin{bmatrix} b_x \\ 0 \\ 0 \end{bmatrix} = \frac{\partial b_x}{\partial x} = 1$		
c)	$\left[\frac{\partial}{\partial x} \hat{i}, \frac{\partial}{\partial y} \hat{j}, \frac{\partial}{\partial z} \hat{k} \right] \begin{bmatrix} b_y \\ b_x \\ 0 \end{bmatrix} = \text{yellow square}$		
d)	$\left[\frac{\partial}{\partial x} \hat{i}, \frac{\partial}{\partial y} \hat{j}, \frac{\partial}{\partial z} \hat{k} \right] \begin{bmatrix} -b_y \\ b_x \\ 0 \end{bmatrix} = 0$		

p. 823 #

a) $\mathbf{r} = x\hat{i} + y\hat{j} + z\hat{k}$

$$r = \sqrt{x^2 + y^2 + z^2}$$

$$\frac{\partial r}{\partial x} = \frac{x}{r}$$

$$\nabla \frac{1}{r} = \left(\frac{\partial}{\partial x} \hat{i}, \frac{\partial}{\partial y} \hat{j}, \frac{\partial}{\partial z} \hat{k} \right) \frac{1}{r}$$

$$= \frac{1}{r^2} \left(\frac{x\hat{i} + y\hat{j} + z\hat{k}}{r} \right) = -\frac{\mathbf{r}}{r^3}$$

Self. qm